



Capstone Project: Comprehensive SOC Incident Response

1. Executive Summary

A simulated cyberattack was conducted against a vulnerable Metasploitable2 system using a Samba usermap exploit. The attack resulted in unauthorized remote access. Detection was successfully performed by Wazuh, followed by triage in TheHive and containment using CrowdSec. MITRE Caldera emulation validated detection coverage for **T1210 (Exploitation of Remote Services)**. The incident was contained efficiently with minimal impact. Key metrics include **MTTD: 2 hours**, **MTTR: 4 hours**, and **Dwell Time: 6 hours**. While detection and response were effective, improvements are needed in reducing dwell time and false positives.

2. Incident Timeline

Time	Event	Details
16:00	Attack Initiated	Metasploit Samba exploit launched
18:00	Detection	Wazuh alert triggered
18:10	Triage	Alert analyzed in TheHive
19:00	Containment	VM isolated, attacker IP blocked
20:00	Resolution	System secured, access removed

3. Attack Simulation

- Tool: Metasploit
- Exploit: exploit/multi/samba/usermap_script
- Target: Metasploitable2
- Outcome: Remote shell access obtained

4. Adversary Emulation (MITRE Caldera)

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 16:00:00	10.0.2.3	Samba exploit detected	T1210



5. Detection and Triage

Wazuh detected suspicious activity and generated an alert. The alert was forwarded to TheHive, where it was classified as high severity and triaged by a SOC analyst.

6. Response and Containment

- Infected VM isolated from the network
 - Attacker IP (10.0.2.3) blocked using CrowdSec
 - Verification: Ping test confirmed attacker unreachable
-

7. SOAR Automation

A TheHive playbook was executed to automate IP blocking via CrowdSec API. The case was automatically created, and response actions completed successfully.

8. Metrics Reporting

- **MTTD:** 2 hours
 - **MTTR:** 4 hours
 - **Dwell Time:** 6 hours
 - **False Positive Rate:** 12%
-

9. Root Cause Analysis (5 Whys)

1. Why did the attack succeed? → Vulnerable Samba service
2. Why was it vulnerable? → Missing patches
3. Why were patches missing? → No patch management process
4. Why no process? → Lack of asset visibility
5. Why lack of visibility? → Insufficient monitoring

Root Cause: Weak vulnerability and asset management practices

10. Recommendations

- Implement patch and vulnerability management program
 - Improve log collection and endpoint visibility
 - Tune detection rules to reduce false positives
 - Deploy SOAR automation for faster response
-



- Conduct regular security awareness training
-

11. Stakeholder Briefing (150 words)

A simulated cyberattack targeting a system vulnerability was successfully detected and contained by the security team. The attacker exploited a known weakness in a network service, gaining unauthorized access. However, our monitoring tools identified the activity within 2 hours, and the threat was fully contained within 4 hours by isolating the affected system and blocking the attacker.

The total exposure time was approximately 6 hours, with no evidence of data loss or lateral movement. The response process worked effectively, supported by automation tools that reduced manual effort.

Analysis shows the root cause was an unpatched system and limited visibility into assets. To prevent similar incidents, we recommend improving patch management, enhancing monitoring coverage, and increasing automation in response actions.

Overall, the organization's security posture remains strong, but these improvements will further reduce risk and response time.